Attendance Register

Date	Day	Activity	Trainer Name & Signature	Trainee Name	Trainee Signature	Comments/Remarks
	Week 1 - Day 1	Introduction & Overview		Abenathi J		
	Week 1 - Day 2	Review of Procedures		Abenathi J		
	Week 1 - Day 3	Technical Assessment – Theory		Abenathi J		
	Week 1 - Day 4	Technical Assessment – Practical		Abenathi J		
	Week 1 - Day 5	Feedback Session		Abenathi J		
	Week 2 - Day 1	Witnessing – Preparation		Abenathi J		
	Week 2 - Day 2	Witnessing – Practical Observation		Abenathi J		
	Week 2 - Day 3	Witnessing – Review		Abenathi J		
	Week 2 - Day 4	Corrective Action & Remediation		Abenathi J		
	Week 2 - Day 5	Final Competency Sign-off		Abenathi J		

	PERSONNEL COMPETENCE DECLARATION FORM		
AzurESky	Document no: ASMF6.1 FORM		
	Compiled by:	Sifiso Nyambi - Laboratory Manager	
	Approved by:	Nonhlanhla Msibi - Operations Manager	

ASMF 6.1 Form

PERSONNEL COMPETENCY DECLARATION FORM

PERSONNEL

Employee name: Jama Abdrathi

ID Number: 02118 0424 080

Coy Number: 4

I declare that I am competent, trained, and authorized to perform my assigned tasks in accordance with the company's management system. I understand and agree to follow all applicable instructions as per the outlined procedure.

Tasks:

Quick ash ☑

ISO Volatile matter ☑

ISO Inherent Moisture ☑

ISO Calorific Value ☑

Total Sulphur ☑

Sample Preparation ☑

Employee signature

AUTHORIZING SIGNATORY

Signatory name: Nonhlanhla Msibi

Designation: Operations Manager

Coy number:

I declare that the above-named individual has been assessed and is authorized to perform the specified tasks in accordance with the company's management system and competency requirements.

Authorizing Signature

Effective date	Rev status	Next review date
07/2025	00	07/2028

sanas South African National Accreditation System

WITNESSING OF ACTIVITY
(PLEASE COMPLETE ONE FORM PER WITNESS)

SANAS Accreditation No/s.

NOTE: When evaluating a new technical signatory applicant, please also complete paragraph 15.

1. Name of Conformity Assessment Body (CAB)	Agrekky (Pty) Ltd.
2. Field / Area of Accreditation	Calorific Value
3. Name of Assessor	Sifiso Nyambi **Date** 18/08/2025

4. **Identification of the specific method / procedure / operation etc. witnessed**
(as described on accreditation schedule) and brief description of method / procedure / operation evaluated.

ISO 1928 Determination of Gross calorific Value.

5. Name of Technical Signatory / Applicant Technical Signatory witnessed	
6. Name of other personnel witnessed	Mbenathu Tania.

7. **Detailed comments, observations and/or reference to assessor's notes. Indicate what was demonstrated and/or talked through as applicable.**

- Clean the bomb electrodes and ports using sandpaper, water, liquid soap and damp towel
- Replace firing wiring and replace damaged orings
- Weigh sample whose mass (0.5000 - 0.5099 g).
- Log into the instrumentation
- Load oxygen into bomb pressure ~2506kpa.
- Put bomb into calorimeter and run 2 checks and 2 controls.
- Plot control chart and then analyse samples.

Calibration

- Run benzoic 5x
- Take closest 3 (min) results
- Press F8 o
- Ensure Ad <0.2
- Analyse benzoic
- Run control

Only calibrate when there is bias on CRM chart

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Additional / General Comments *This space may also be used to expand on comments in specific sections*

Con 1: 24.036
Con 2: 24.049

1493 ⇒ 18.469
1493 ⇒ 18.469
} Average = 18.47 MJ/kg

8. Comments on internal control procedures witnessed, and acceptability of the outcome (as applicable)

⇒ Diff b/w replicate results < 0.12 MJ/kg
⇒ Procedure to follow present.

9. Comments on the traceability of reference standards / materials, and the calibration and/or maintenance of measuring equipment (as applicable)

⇒ Benzoic acid tablets used for calibration. Units given as MJ/kg, traceable to SI unit.
⇒ CRM AS002 used to check for drifts.

©SANAS

10. Comments on uncertainty of measurement and validation data for the method / procedure as applicable)

(Not applicable to Inspection Bodies)

* Budget of uncertainty available.
* Uncertainties associated with CRM are present in CoA.

11. Comments on training, qualifications, competency, and authorisation records of the personnel member witnessed, and the availability of a current job description (however named)

⇒ Diploma in Chemistry.
⇒ Job description present.

12. Facilities and Environmental conditions (where applicable)

⇒ Room temp is within acceptability criteria of $19^{\circ}C \leq T \leq 26^{\circ}C$
⇒ Housekeeping done and room is clean & tidy.

13. Personnel Registration / Certification if applicable to the scope of accreditation:

(Please provide detailed comments)

14. Recommendation / conclusions of the assessor with regards to competence of the person and the method/ procedure / operation witnessed

⇒ Analyst follows procedure and understand the principle behind analysis. Can reason and make reflective thoughts

⇒ Analyst is declared competent

Assigned value = 18.84
Obtained value = 18.47

Signature of Technical Assessor / Expert		Signed by Team Leader	

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15. Evaluation of *NEW* Technical Signatory Applicant (If applicable)

15.1 Applicant competence declared by the CAB:
(Please provide detailed comments)

15.2 Level of understanding of the objectives and knowledge of the method / procedure / operation:
(Please provide detailed comments)

15.3 Ability to critically evaluate and interpret results/reports:
(Please provide detailed comments)

15.4 Does the applicant accept responsibility for the content and validity of results / reports he / she will be signing / authorising (refer to Act 19 of 2006)?
(Please provide detailed comments)

15.5 Is the applicant conversant with the management system implemented within the CAB:
(Please provide detailed comments)

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15.6 Level of understanding of SANAS and the accreditation requirements (Refer to F147 -Terms and Conditions of Accreditation and R03 - Nominated Representative and Signatories: Responsibilities, Qualifications and Approval, and Accreditation of Conformity Assessment, Calibration and Good Laboratory Practices (Act 19 of 2006)):

(Please provide detailed comments)

15.7 Assessment team recommendation in terms of approval as a Technical Signatory:

Recommended ☐

Please list the scope/methods/procedures etc. recommended for:

NOT Recommended ☐

Please provide reasons:

Signature of Technical Assessor / Expert		**Signed by Team Leader**	

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WITNESSING OF ACTIVITY
(PLEASE COMPLETE ONE FORM PER WITNESS)

SANAS Accreditation No/s.

NOTE: *When evaluating a new technical signatory applicant, please also complete paragraph 15.*

1. Name of Conformity Assessment Body (CAB)	Azurisky (Pty) Ltd.
2. Field / Area of Accreditation	ISO Volatile Matter
3. Name of Assessor	Ifiso Nyambi — Date: 21/08/2025
4. Identification of the specific method / procedure / operation etc. witnessed (as described on accreditation schedule) and brief description of method / procedure / operation evaluated.	ISO 562 – Coke Coal: Determination of Volatile Matter in coal.
5. Name of Technical Signatory / Applicant Technical Signatory witnessed	
6. Name of other personnel witnessed	Mbenathi Jama.

7. Detailed comments, observations and/or reference to assessor's notes. Indicate what was demonstrated and/or talked through as applicable.

- Look at environmental conditions: $19° < T < 26°C$, RH: 25–45%
- Balance verification (check for bubble ⊙)
- Physical inspect crucibles, check for cracks/chips.
- Use furnace to clean them (i.e. insert empty crucible for 5 min)
- Balance cleanliness.
- Furnace verifications done on Mondays.
- Weigh crucibles and samples as per worksheet.
- All batches must be accompanied by CRM.
- Limits: 29.41 UL, 27.69 LL.
- Insert crucible into balance and take the reading.
- Control 1: Mass crucible = 12.3708g, sample mass = 0.9819g
  2: Mass crucible = 12.8955, sample mass = 1.0108g
- 1493 1: Mass crucible = 13.2653g, sample mass = 1.0076g
  2: Mass crucible = 12.4272, sample mass = 0.9926g

Can you use unlabelled crucibles? No, it makes it easier to remember which crucible had what sample.

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Additional / General Comments *This space may also be used to expand on comments in specific sections*

⇒ Place Aand onto a tray and proceed to the furnace room.
⇒ Use AS015TF form to record initial temperature (908C)
⇒ Insert the crucibles into furnace & start the timer.
⇒ At 4 min (T) = 898°C
At 7 min (T) = 900°C
⇒ Cool the Aand on the bench for = 5 min.
⇒ Cool the Aand in dessicator for another = 10 min
⇒ 1493, 1: 26,52
2: 26,57
} Average = 26.55

Furnace ID = AS01338

8. Comments on internal control procedures witnessed, and acceptability of the outcome (as applicable)

* Use of CRM to check for drifts

9. Comments on the traceability of reference standards / materials, and the calibration and/or maintenance of measuring equipment (as applicable)

⇒ All equipments (balance, furnace, mass piece etc) are calibrated
⇒ CRMLOD2 used to check for drifts!

10. Comments on uncertainty of measurement and validation data for the method / procedure as applicable)

(Not applicable to Inspection Bodies)

11. Comments on training, qualifications, competency, and authorisation records of the personnel member witnessed, and the availability of a current job description (however named)

- Diploma in Chemistry.
- Job description present.

12. Facilities and Environmental conditions (where applicable)

- Temp currently at 22.4°C
- Toplocher cleaned to avoid contamination

13. Personnel Registration / Certification if applicable to the scope of accreditation:
(Please provide detailed comments)

14. Recommendation / conclusions of the assessor with regards to competence of the person and the method/ procedure / operation witnessed

- Analyst competent

Signature of Technical Assessor / Expert		**Signed by Team Leader**	

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15. Evaluation of *NEW* Technical Signatory Applicant (If applicable)
15.1 Applicant competence declared by the CAB: *(Please provide detailed comments)*
15.2 Level of understanding of the objectives and knowledge of the method / procedure / operation: *(Please provide detailed comments)*
15.3 Ability to critically evaluate and interpret results/reports: *(Please provide detailed comments)*
15.4 Does the applicant accept responsibility for the content and validity of results / reports he / she will be signing / authorising (refer to Act 19 of 2006)? *(Please provide detailed comments)*
15.5 Is the applicant conversant with the management system implemented within the CAB: *(Please provide detailed comments)*

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15.6 Level of understanding of SANAS and the accreditation requirements (Refer to F147 -Terms and Conditions of Accreditation and R03 - Nominated Representative and Signatories: Responsibilities, Qualifications and Approval, and Accreditation of Conformity Assessment, Calibration and Good Laboratory Practices (Act 19 of 2006)):

(Please provide detailed comments)

15.7 <u>Assessment team recommendation in terms of approval as a Technical Signatory:</u>

Recommended ☐

Please list the scope/methods/procedures etc. recommended for:

NOT Recommended ☐

Please provide reasons:

Signature of Technical Assessor / Expert		**Signed by Team Leader**	

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WITNESSING OF ACTIVITY

(PLEASE COMPLETE ONE FORM PER WITNESS)

SANAS Accreditation No/s.

NOTE: When evaluating a new technical signatory applicant, please also complete paragraph 15.

1. Name of Conformity Assessment Body (CAB)	Mzuri Sty (Pty) Ltd.		
2. Field / Area of Accreditation	~~Iso Volatile~~ SB. Q-lab.		
3. Name of Assessor	Ofiso Myambi	Date	18/08/2025

4. **Identification of the specific method / procedure / operation etc. witnessed**
(as described on accreditation schedule) and brief description of method / procedure / operation evaluated.

~~Iso Volatile Matter 562~~ SB
Q-lab

5. Name of Technical Signatory / Applicant Technical Signatory witnessed	
6. Name of other personnel witnessed	Menathi Jama -

7. **Detailed comments, observations and/or reference to assessor's notes. Indicate what was demonstrated and/or talked through as applicable.**

- Check environmental temp : should be $19^{\circ}C < T < 26^{\circ}C$, and $25\% < H < 45\%$
  If temp is out of spec, use AC to fix it.
- Balance verification.
- Clean crucibles, clean spatula.
- Condition ; "samples must adapt to lab environment!"
- $0,2495 < mass < 0.5000\,g$
- Insects control and understands the purpose of it.
- control: Cno 6 : 30.8374g, mass = 0.2814g
  Cno 7 : 37.2624g, mass = 0.3117g
  1493 : Cno 16 : 38.0249g, mass = 0.2947g
  1493, Cno 17 : 33.7562g, mass = 0.3082g

  } cleans balance if there's a spillage.
- Check temp of furnace ($815^{\circ}C$)
- Set timer for an hour. 11h27
- Fill in AS014FT form (QA workbook)
- Cool samples for 10 min

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Additional / General Comments *This space may also be used to expand on comments in specific sections*

Control 1: 22.17 } 22.30 Within limits.
2: 22.43 }

1493, 1: 32.03 } 32.25
1 2: 32.48 }

8. Comments on internal control procedures witnessed, and acceptability of the outcome (as applicable)

- Control used and is within limits.
- Procedure followed.

9. Comments on the traceability of reference standards / materials, and the calibration and/or maintenance of measuring equipment (as applicable)

CRM 005 used for competency assessment.
Assigned value = 32.79
Absolute diff = $|32.25 - 32.79| = 0.54\%$

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10. Comments on uncertainty of measurement and validation data for the method / procedure as applicable)

(Not applicable to Inspection Bodies)

11. Comments on training, qualifications, competency, and authorisation records of the personnel member witnessed, and the availability of a current job description (however named)

⇒ JD present.
⇒ Diploma in chemistry.

12. Facilities and Environmental conditions (where applicable)

$T_{initial} = 21.4^{\circ}C$
$T_{final} = 22.3^{\circ}C$
$RH = 46\%$

13. Personnel Registration / Certification if applicable to the scope of accreditation:
(Please provide detailed comments)

14. Recommendation / conclusions of the assessor with regards to competence of the person and the method/ procedure / operation witnessed

⇒ Candidate is competent
⇒ Q-ash assigned value = 32.79% (AD)
Obtain value = 32.25%
Since diff b/w AV < 2%.

Signature of Technical Assessor / Expert		**Signed by Team Leader**	

©SANAS

15. Evaluation of *NEW* Technical Signatory Applicant (If applicable)

15.1 Applicant competence declared by the CAB:
(Please provide detailed comments)

15.2 Level of understanding of the objectives and knowledge of the method / procedure / operation:
(Please provide detailed comments)

15.3 Ability to critically evaluate and interpret results/reports:
(Please provide detailed comments)

15.4 Does the applicant accept responsibility for the content and validity of results / reports he / she will be signing / authorising (refer to Act 19 of 2006)?
(Please provide detailed comments)

15.5 Is the applicant conversant with the management system implemented within the CAB:
(Please provide detailed comments)

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15.6 Level of understanding of SANAS and the accreditation requirements (Refer to F147 -Terms and Conditions of Accreditation and R03 - Nominated Representative and Signatories: Responsibilities, Qualifications and Approval, and Accreditation of Conformity Assessment, Calibration and Good Laboratory Practices (Act 19 of 2006)):

(Please provide detailed comments)

15.7 Assessment team recommendation in terms of approval as a Technical Signatory:

Recommended ☐

Please list the scope/methods/procedures etc. recommended for:

NOT Recommended ☐

Please provide reasons:

Signature of Technical Assessor / Expert		**Signed by Team Leader**	

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WITNESSING OF ACTIVITY
(PLEASE COMPLETE ONE FORM PER WITNESS)

SANAS Accreditation No/s.

NOTE: When evaluating a new technical signatory applicant, please also complete paragraph 15.

1. Name of Conformity Assessment Body (CAB)	AZURESKY QTM LTD		
2. Field / Area of Accreditation	LABORATORY		
3. Name of Assessor	MAPONYA TMJ	Date	22/08/2025

4. Identification of the specific method / procedure / operation etc. witnessed
(as described on accreditation schedule) and brief description of method / procedure / operation evaluated.

Iso Moisture as per procedure

5. Name of Technical Signatory / Applicant Technical Signatory witnessed	
6. Name of other personnel witnessed	ABENATHI JAMA

7. Detailed comments, observations and/or reference to assessor's notes.
Indicate what was demonstrated and/or talked through as applicable.

What is inherent moisture
> Quantity of moisture contained within coal
> Mentions the checking of the room temperature to be within 19°C - 26°C
> Performs balance verification
> Was guided to check the validity of the calibration stickers on the instrument.
> Crucible mass - 31,27189 1,055 } Control
30,7805 0,9891
Crucmass - 31,6477 0,9891
- 29,4491 0,9739
The control failed repeatability criteria, thereby rendering the samples null. because there is no indicator to show that the analyses was reliable

$\frac{100 - Im}{100 - Sm} \times Analysis$

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AD - AR

Additional / General Comments *This space may also be used to expand on comments in specific sections*

Control:
Crucible 1 – 2.82
2 – 2.84

Crucible: 3 – 4.41 } Sample
4 – 4.45

8. Comments on internal control procedures witnessed, and acceptability of the outcome (as applicable)

9. Comments on the traceability of reference standards / materials, and the calibration and/or maintenance of measuring equipment (as applicable)

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10. Comments on uncertainty of measurement and validation data for the method / procedure as applicable)
(Not applicable to Inspection Bodies)
11. Comments on training, qualifications, competency, and authorisation records of the personnel member witnessed, and the availability of a current job description (however named)
12. Facilities and Environmental conditions (where applicable)
13. Personnel Registration / Certification if applicable to the scope of accreditation: *(Please provide detailed comments)*
14. Recommendation / conclusions of the assessor with regards to competence of the person and the method/ procedure / operation witnessed

Signature of Technical Assessor / Expert		**Signed by Team Leader**	

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15. Evaluation of *NEW* Technical Signatory Applicant (If applicable)

15.1 Applicant competence declared by the CAB:
(Please provide detailed comments)

15.2 Level of understanding of the objectives and knowledge of the method / procedure / operation:
(Please provide detailed comments)

15.3 Ability to critically evaluate and interpret results/reports:
(Please provide detailed comments)

15.4 Does the applicant accept responsibility for the content and validity of results / reports he / she will be signing / authorising (refer to Act 19 of 2006)?
(Please provide detailed comments)

15.5 Is the applicant conversant with the management system implemented within the CAB:
(Please provide detailed comments)

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15.6 Level of understanding of SANAS and the accreditation requirements (Refer to F147 -Terms and Conditions of Accreditation and R03 - Nominated Representative and Signatories: Responsibilities, Qualifications and Approval, and Accreditation of Conformity Assessment, Calibration and Good Laboratory Practices (Act 19 of 2006)):

(Please provide detailed comments)

15.7 Assessment team recommendation in terms of approval as a Technical Signatory:

Recommended ☐

Please list the scope/methods/procedures etc. recommended for:

NOT Recommended ☐

Please provide reasons:

Signature of Technical Assessor / Expert		**Signed by Team Leader**	

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AzurESky "ensure quality of test"	AZUREKY PTY LTD	Document # AZCA-003	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF CALORIFIC VALUE IN COAL		

PERSONAL DETAILS:

Name:	Abenathi Jama
Staff No:	
Job Title:	Analyst

ASSESSMENT:

Attempt:	1st
Date:	14 August 2025
Time:	14:52

ASSESSORS:

Name & Staff nr.:	1.	Ifiso Nyambi
	2.	
	3.	
	4.	
Observers: (optional)	1.	2.
Translator: (if required)		

Instructions and Information

1.	The competency-based assessment instrument was developed in line with the requirements to be declared competent in **Determination of the Calorific Value in Coal** **Desired Outcome:** At the end of this module the employee should be able to perform total sulphur determination in line with the test procedure in a manner which will give accurate, repeatable and traceable results. The employee should have adequate knowledge of the method and its limitations
2.	The competency and development tool is based on ISO 1928
3.	You are allowed to use the following documentation during the assessment: • Any manual(s) associated with the Calorific Value determination(s) in use. • Any Corporate Standard, Corporate Guideline or Corporate Manual. • Any other relevant **printed** reference material you may feel will assist you.
4.	You are going to demonstrate the determination of Calorific Value. During the assessment, the assessor may ask you additional relevant questions, the answer to which he/she will then immediately note down on the assessment sheet. The assessment consists of **23** points of which **14** are critical. To be declared competent, you must obtain 80% for the assessment. You are not allowed to get one of the critical questions wrong. The assessment shall then be stopped, and the result will be that you are not yet competent. Some of the questions may be structured to assess your reflective competence. There is a set time limit of **60** minutes for the completion of the assessment.
5.	You have the right to an interpreter and the right to have an observer present during the assessment.
6.	You are allowed to ask the assessor(s) to clarify any question during the assessment.

AzurESky "ensure quality of test"	AZUREKY PTY LTD	Document # AZCA-003	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF CALORIFIC VALUE IN COAL		

Assessment of Calorific Value

100%

Date: 14/08/25

Candidate name: ~~Abarathi~~ **Staff No:** _____

Candidate Signature: _____

No	Question:	Points	Score
1.	What is meant by calorific Value?	*	
	Energy output generated by coal sample during combustion	1	1
2.	What chemical is used to calibrate the machine?	*	
	benzoic acid tablet	1	1
3.	What is meant by repeatability?		
	Analyses done by same operator under same condition	1	1
4.	What is used to clean the bomb vessel inside?		
	A slightly damp towel	1	1
5.	Can grease or oil be used on oxygen gauges?	1	1
	no can not		
6.	Complete the sentence: Duplicate results by the same Analyst using the same bomb under same conditions	* 2	2
7.	Which documents must be used to record raw data?	* 1	1
	AS005TF-CV Calibration form		
8.	How often is the instrument calibrated?	1	1
	It only done when there is a blast on the Control Chart and have exhausted all possible options.		

9.	How often are and with what are drift checks analyzed? After every 10 runs drift checks are analyzed with control	* 2	2
10.	Can the bombs be immersed in water? no, Can not	1	1
11.	A Benzoic Acid can be used until the bottle is empty True or False? False	* 1	1
12.	Stanley knife is used to cut o-rings. True or False? True	1	1
13.	Within which masses are test samples controlled 0.5000 - 0.5999g	* 2	2
14.	What is the repeatability limit for Calorific Value determination? <0.12 MJ/kg	* 1	1
15.	When is the maintenance on Calorific Value done? each time CV analyses is done	* 1	1
16.	How many samples are used in repeat analyses? duplicate	* 1	1
17.	Who plots the control charts? Analyst	* 1	1
18.	What material is used to clean electrodes? Sandpaper	1	1
19.	Calorific value crucibles can never be reused True/ False False	1	1
20.	What gas is used in the calorific value analyzer?	1	1

	Oxygen		
	Total:	23	23

Comments by Assessor(s):

Comments by Candidate:

Assessor Declaration:

I hereby state that the candidate has been assessed according to the requirements of this document and found to be *competent / not yet competent** **and** *may / may not* proceed* **to the Assessment Panel for authorisation.**

Assessor Name Tiso Nyamb **Signature:** _____ **Date:** 14/08/2025

Learner Signature: _____ **Date:** _____

Note: 1. This shall be done after theoretical test has been completed.
2. Trainee to be constantly supervised until Initial training is signed off by competent trainer.

Sample Used	Lims Number	Known Value	Duplicate Test Results	
AD CRM 005	1493	18.84	18.469	18.469
DB	1493	19.67	19.33	
Acceptance Tolerance: 0.12% between duplicate results				

AzurESky	AZURESKY PTY LTD	Document # AZCA-005	Revision# 0
Subject	ASSESSMENT FOR SAMPLE PREPARATION		

PERSONAL DETAILS:

Name:	Abenathi Jama
Job Title:	Analyst

ASSESSMENT:

Attempt:	1st
Date:	13/05/25
Time:	

ASSESSORS:

Name	1.	Sifiso M.
	2.	MAPONYA TMJ

Instructions and Information

1.	The competency and development tool is based on the Sample Preparation **Desired Outcome:** At the end of this module the employee should be able to perform the determination of ash content in line with the test procedure in a manner which will give accurate, repeatable and traceable results. The employee should have adequate knowledge of the method and its limitations
2.	You are allowed to use the following documentation during the assessment: • Any other relevant **printed** reference material you may feel will assist you.
3.	You are going to demonstrate the determination of the Sample Preparation. During the assessment, the assessor may ask you additional relevant questions, the answer to which he/she will then immediately note down on the assessment sheet. The assessment consists of **20** points of which **2** are critical. To be declared competent, you must obtain 80% for the assessment. You are not allowed to get one of the critical questions wrong. The assessment shall then be stopped, and the result will be that you are not yet competent. Some of the questions may be structured to assess your reflective competence.
4.	You have the right to an interpreter and the right to have an observer present during the assessment.
5.	You are allowed to ask the assessor(s) to clarify any question during the assessment.

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-005	Revision# 0
Subject	ASSESSMENT FOR SAMPLE PREPARATION		

Assessment Sample Preparation

Date: 13/08

Candidate name: Abenathi **Staff No:** _____

Candidate Signature: _____

95%

No	Question:	Points	Score
1.	Name two apparatus used to mix sample for preparation Homogenizer / Crusher / Splitter	2	2
2.	What is the purpose of crushing the sample? Area To increase surface of the coal and in a way homogenust sample	2	2
3.	Why do we air dry the sample if it is visible wet? When crushing & pulverizing So that it will not easier to work with coal	2	2
4.	Why is housekeeping such an important part of Good Laboratory Practices (GLP)? Safety risk to operators It can prevent contamination and	1	1
5.	Name 5 steps to be followed air-drying the sample Crush sample from 25mm, 10mm and 3mm Crusher Air-dry sample if it wet Pulverize sample Homogenise Split the sample	* 5	5
6.	At what temperature is the test sample dry for the general analysis? 40°C	* 1	1
7.	When is the pulveriser verification to be done? Each time pulveriser is used	1	1
8.	What mass is extracted from the homogeneity drum during packaging? ~~600g~~ ~~600g~~ 600g	1	1
9.	What are the acceptance criteria for crusher efficiency? 90% and above	1	1

10.	Samples should be stored into a non-labeled bottle (True or false) Support your answer. labeled Pelet, it will be difficult to identify if it not	2	2
11.	Reference samples are kept in the lab for future disputes by the clients. (True or false) false ✓	1	
12.	How frequently must stored samples be checked for stability? once a month	1	1
	Total:	20	19

Comments by Assessor(s):

Comments by Candidate:

Assessor Declaration:

I hereby state that the candidate has been assessed according to the requirements of this document and found to be *competent / not yet competent** **and** *may / may not* proceed* **to the Assessment Panel for authorisation.**

* Delete as applicable.

Assessor Name [illegible] **Signature:** [illegible] **Date:** 13/08/25

If you do not agree with the feedback, please discuss this. If you can not come to an agreement, ask for an explanation of the appeals procedure.

I declare that I am satisfied with this assessment and can perform these tasks.

Learner Signature: [illegible] Date: 13/08/25

Senior analyst Signature: _____ Date: _____

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-002	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF VOLATILE MATTER		

PERSONAL DETAILS:

Name:	Abenathi Jama
Staff No.:	
Job Title:	Analyst

ASSESSMENT:

Attempt:	1st
Date:	14 August 2025
Time:	13H59

ASSESSORS:

Name & Staff nr.:	1.	Difiso H (02)
	2.	
	3.	
	4.	
Observers: (optional)	1.	2.
Translator: (if required)		

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-002	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF VOLATILE MATTER		

Instructions and Information

1.	The competency-based assessment instrument was developed in line with the requirements to be declared competent in **Determination of Volatile Matter in Coal** **Desired Outcome:** At the end of this module the employee should be able to perform volatile matter determination in line with the test procedure in a manner which will give accurate, repeatable and traceable results. The employee should have adequate knowledge of the method and its limitations
2.	The competency and development tool is based on ISO 562
3.	You are allowed to use the following documentation during the assessment: • Any manual(s) associated with the Volatile Matter determination(s) in use. • Any Corporate Standard, Corporate Guideline or Corporate Manual. • Any other relevant **printed** reference material you may feel will assist you.
4.	You are going to demonstrate the use of determination of the Volatile Matter. During the assessment, the assessor may ask you additional relevant questions, the answer to which he/she will then immediately note down on the assessment sheet. The assessment consists of **27** points of which **24** are critical. To be declared competent, you must obtain 80% for the assessment. You are not allowed to get one of the critical questions wrong. The assessment shall then be stopped, and the result will be that you are not yet competent. Some of the questions may be structured to assess your reflective competence. There is a set time limit of **60** minutes for the completion of the assessment
5.	You have the right to an interpreter and the right to have an observer present during the assessment.
6.	You are allowed to ask the assessor(s) to clarify any question during the assessment.

Assessment for Volatile Matter

98 6/6

Date: 14 / 08

Candidate name: ~~Abenath~~ **Staff No:** _____

Candidate Signature: ~~Jeune~~

No	Question:	Points	Score
1.	At what temperature is the test sample incinerated?	* 1	1
	900 ± 5°C ✓		
2.	What does one use to monitor the time in the furnace?	* 1	1
	Stop watch ✓		
3.	What is the maximum time for incineration of the test sample?	* 1	1
	7 Minute ± 2 secs. ✓		
4.	To what temperature must the volatile furnace recover after 4 minutes?	* 1	1
	896°C ✓		
5.	Where does one cool the crucible and stands?	1	1
	Work bench & desiccator. ✓		
6.	Complete the sentence: If the furnace has not recovered fail / reject the result and run Repeat analysis ✓	* 2	2
7.	What is meant by repeatability?	* 3	3
	Analysis done by ~~one~~ same operator, using same apparatus under same conditions ✓		
8.	How often drifts are checks run?	* 1	

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-002	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF VOLATILE MATTER		

	After every 4 samples		1
9.	How often and by whom are the verification checks done? (Monday)	* 2	2
	once a week by analyst /operator		
10.	Between which masses are the masses for volatile determination controlled?	* 1	1
	0.95 - 1.05g		
11.	The insertion time of a volatile stand can be 10 seconds. True or False?	* 1	1
	false		
12.	A volatile crucible stand can be kept in the furnace for 7min± 8 seconds. True or False?	* 1	1
	false		
13.	How does one identify volatile crucible and lids? on both crucible & lid	* 2	2
	label them with a corresponding number.		
14.	What is the repeatability limit for volatile determination?	* 1	1/2
	0.3% for volatile matter less than 10%		
15.	What is subtracted after testing to get volatile results?	* 1	1
	Inherent moisture ~~matter~~ water		
16.	How many samples must be used in the repeat analyses?	1	1
	duplicate		
17.	How often are crucibles cleaned?	* 1	1
	After each test		
18.	What should the distance between the crucible and the lid?	1	1
	0.5mm		
19.	How does one ensure that new lids fit their crucible? no more than 0.5mm distance between crucible & lid	* 1	1
	Make sure that there is no space between.		
20.	What laboratory documents are to be used when	*	

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-002	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF VOLATILE MATTER		

	determining the Volatile Matter Temperature Recovery	1	1
21	Are you allowed to carry out volatile analysis with vacant places in your stand?	* 2	2
	Not allowed, Stand should have 4-		
	Total:	27	26.5

crucible to allow even distribution of heat

Comments by Assessor(s):

Comments by Candidate:

Assessor Declaration:

I hereby state that the candidate has been assessed according to the requirements of this document and found to be *competent / not yet competent** **and** *may / may not* proceed* **to the Assessment Panel for authorisation.**

Assessor Name Tafiso M **Signature:** _____ **Date:** 14/08/2025

Learner Signature: _____ **Date:** 14/08/25

Note: 1. This shall be done after theoretical test has been completed.
2. Trainee to be constantly supervised until Initial training is signed off by competent trainer.

Sample Used	Lims Number	Known Value	Duplicate Test Results	
CRM 005	1493	22.09	22.11	22.12
Dry base	1493	23.06		23.15
Acceptance Tolerance: 0.3% or 4% of average between duplicate results				

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-004	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF THE ASH CONTENT (COAL-QUICK ASH METHOD)		

PERSONAL DETAILS:

Name:	Athenathi Jama
Job Title:	Analyst

ASSESSMENT:

Attempt:	1st.
Date:	14/08/2025.
Time:	-

ASSESSORS:

Name	1.	Afiso Nyambi
	2.	

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-004	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF THE ASH CONTENT (COAL-QUICK ASH METHOD)		

Instructions and Information

1.	The competency and development tool is based on the ISO 1171 **Desired Outcome:** At the end of this module the employee should be able to perform the determination of ash content in line with the test procedure in a manner which will give accurate, repeatable and traceable results. The employee should have adequate knowledge of the method and its limitations
2.	You are allowed to use the following documentation during the assessment: • Any manual(s) associated with the Ash Content determination(s) in use. • Any Corporate Standard, Corporate Guideline or Corporate Manual. • Any other relevant **printed** reference material you may feel will assist you.
3.	You are going to demonstrate the determination of the Ash Content. During the assessment, the assessor may ask you additional relevant questions, the answer to which he/she will then immediately note down on the assessment sheet. The assessment consists of **23** points of which **13** are critical. To be declared competent, you must obtain 80% for the assessment. You are not allowed to get one of the critical questions wrong. The assessment shall then be stopped, and the result will be that you are not yet competent. Some of the questions may be structured to assess your reflective competence.
4.	You have the right to an interpreter and the right to have an observer present during the assessment.
5.	You are allowed to ask the assessor(s) to clarify any question during the assessment.

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-004	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF THE ASH CONTENT (COAL-QUICK ASH METHOD)		

Assessment Ash Content

100%

Date: 14/05

Candidate name: Ayanthi **Staff No:** _____

Candidate Signature: ~~Ayanthi~~

No	Question:	Points	Score
1.	At what temperature is the ash sample incinerated to constant mass?	* 1	1
	815°C ±10		
2.	What problems could be encountered if the sample is incinerated in a furnace without a specified ramping profile?	1	1
	Less Sulfur Content will be retained		
3.	Between which masses are the ash samples controlled?	* 2	2
	0.2495 – 0.5g		
4.	What is the maximum temperature at which samples can be inserted into the furnace?	2	2
	815°C		
5.	Complete the sentence: The sample is incinerated at 815°C throughout analysis for a period of 1 hour and if sample has not completely ashed ~~if 30 minutes~~ After hour an additional 30 minutes can be added.	* 2	2
6.	Where are the crucibles cooled after the analysis is complete?	* 2	2
	Sample are cooled work bench, right next to the furnace for 10-15 minutes		
7.	What accompanies each batch of samples?	1	1
	Control		
8.	What is the maximum number of crucibles allowed in a furnace?	* 1	1
	22		

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-004	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF THE ASH CONTENT (COAL-QUICK ASH METHOD)		

9.	Who is responsible for investigating any deviation?	* 1	1
	Analyst		
10.	Two grams of samples can be used for ash. True or False?	1	1
	Falst		
11.	The repeatability of ash is 0.3%. True or False?	1	1
	Falst		
12.	The unit for reporting ash is grams True or False?	* 1	1
	Falst		
13.	What is the formula for calculating the ash content?	1	
	%Ash = $\frac{M_3 - M_1}{M_2} \times 100$		1
14.	Who is responsible for plotting control charts?	* 1	
	Analyst		1
17.	A single repeat sample can be used for ash True/False	* 1	1
	Falst		1
18.	At what temperature are the ash crucibles weighed back?	1	1
	RT – 26°C.		
19.	What is the repeatability limit for ash determinations?	* 1	1
	Ash content < 10% RL of 0.2%; AC > 10% RL 2% (MEAN)		
20.	Unmarked and unidentified crucibles can be used for ash determination. True/ False	* 1	
	Falst		
	Total:	23	

Comments by Assessor(s):

_____

_____

Comments by Candidate:

_____

_____

_____

_____

Assessor Declaration:

I hereby state that the candidate has been assessed according to the requirements of this document and found to be *competent / not yet competent** **and** *may / may not* proceed* **to the Assessment Panel for authorisation.**

* Delete as applicable

Assessor Name Afiso M **Signature:** _____ **Date:** 14/08/2025

If you do not agree with the feedback, please discuss this. If you cannot come to an agreement, ask for an explanation of the appeals procedure.

I declare that I am satisfied with this assessment, and I can perform these tasks.

Learner Signature: _____ Date: _____

Senior analyst/supervisor Signature: _____ Date: _____

Learner Signature: _____ Date: 14/08/25

Note: 1. This shall be done after theoretical test has been completed.
2. Trainee to be constantly supervised until Initial training is signed off by competent trainer.

Sample Used	Lims Number	Known Value	Duplicate Test Results	
CRM 005	1493	32.79	32.03%	32.48%
Dry Basc	1493	34, 33	33, 74	

Acceptance Tolerance: 0.2% between duplicate results for Ash≤10% and 2% of average for ash>10%.

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-001	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF MOISTURE CONTENT OF COAL IN THE ANALYSIS COAL SAMPLE		

PERSONAL DETAILS:

Name:	Abenathi Jama
Job Title:	Analyst

ASSESSMENT:

Attempt:	2nd
Date:	14 August 2025
Time:	15:35

ASSESSORS:

Name	1.	Ayiso Nyambi
	2.	

Instructions and Information

1.	The competency-based assessment instrument was developed in line with the requirements to be declared competent in **the Determination of Inherent Moisture** **Desired Outcome:** At the end of this module the employee should be able to perform Determination of inherent Moisture in line with the test procedure in a manner which will give accurate, repeatable and traceable results. The employee should have adequate knowledge of the method and its limitations
2.	You are allowed to use the following documentation during the assessment: • Any manual(s) associated with Inherent Moisture determination(s) in use. • Any Corporate Standard, Corporate Guideline or Corporate Manual. • Any other relevant **printed** reference material you may feel will assist you.
3.	You are going to demonstrate the determination of the inherent moisture During the assessment, the assessor may ask you additional relevant questions, the answer to which he/she will then immediately note down on the assessment sheet. The assessment consists of **26** points of which **16** are critical (*). To be declared competent, you must obtain 80% for the assessment. You are not allowed to get one of the critical questions wrong. The assessment shall then be stopped, and the result will be that you are not yet competent. Some of the questions may be structured to assess your reflective competence.
4.	You have the right to an interpreter and the right to have an observer present during the assessment.
5.	You are allowed to ask the assessor(s) to clarify any question during the assessment.

AzurESky	AZURESKY PTY LTD	Document # AZCA-001	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF MOISTURE CONTENT OF COAL IN THE ANALYSIS COAL SAMPLE		

Assessment Inherent Moisture

100%

Date: 14/08

Candidate name: Aparthi Staff No: _____

Candidate Signature: _____

No	Question:	Points	Score
1.	At what temperature is the test sample analysed?	* 1	1
	105°C - 110°C		
2.	Who is responsible for plotting control charts?	* 1	1
	Analyst		
3.	Who is responsible for investigating any deviation?	* 1	1
	Analyst		
4.	What is meant by the term repeatability? Analyses done by same analyst using same apparatus under same condition	2	2
5.	What does high inherent moisture content indicate? low rank coal	1	1
6.	Complete the sentence: Oxidized Coals become porous Oxygen enters easily which would lead to combustion, effect calorific value. It like sponge reading to absorb air	* 2	2
7.	Where must the crucibles be cooled and for how long?	* 2	2
	Wait bench and desiccator for 10 minute		
8.	What accompanies each batch of samples for inherent moisture?	* 1	1
	Control		
9.	How must the crucible lid be? the lid must be open and no cracks	1	1

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-001	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF MOISTURE CONTENT OF COAL IN THE ANALYSIS COAL SAMPLE		

10.	Name the one desiccant that can be used?	1	1
	Silica gel		
11.	The inherent moisture sample is incinerated for one hour. True or False?	* 1	1
	False		
12.	The inherent moisture crucible is weighed back while still hot. True or False?	* 1	1
	False		
13.	Inherent moisture crucibles are cooled with the lid open True or False?	* 1	1
	False		
14.	What is the function of silica gel in the dessicators?	1	1
	absorb moisture		
15.	How does one regenerate silica gel?	1	1
	by placing silica gel into the moisture oven		
16.	How many samples must be used in the repeat analyses?	1	1
	duplicate		
17.	Within which limits must mass be controlled?	* 1	1
	0.95 - 1.05g		
18.	What are the repeatability limits for inherent moisture?	* 4	4
	IM Content $\leq 5\%$. RL of 0.10%.		
	IM Content $\& > 5\%$. RL of 0.15%.		
19.	Inherent moisture can only be done when the balance has been verified True/ False	* 1	
	True		
20.	What is the relation between inherent moisture and volatile matter?	1	
	Inherent moisture is needed for correction of volatile matter.		
	Total:	26	26

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-001	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF MOISTURE CONTENT OF COAL IN THE ANALYSIS COAL SAMPLE		

Comments by Assessor(s):

Comments by Candidate:

Assessor Declaration:

I hereby state that the candidate has been assessed according to the requirements of this document and found to be *competent / not yet competent and *may / may not* proceed* to the Assessment Panel for authorization.**

Assessor Name [signature] **Signature:** [signature] **Date:** 14/08/2015

Learner Signature: [signature] Date: 14/08/25

Note: 1. This shall be done after theoretical test has been completed.
2. Trainee to be constantly supervised until Initial training is signed off by competent trainer.

Sample Used		Lims Number	Known Value	Duplicate Test Results	
005		1493	4.201	4.41	4.45
Acceptance Tolerance: 0.15% between duplicate results for IM >5% and 0.1 for IM ≤5%					

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-001	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF MOISTURE CONTENT OF COAL IN THE ANALYSIS COAL SAMPLE		

PERSONAL DETAILS:

Name:	Abenathi Jama
Job Title:	Analyst

ASSESSMENT:

Attempt:	1st
Date:	13 August 2025
Time:	13H05 - 14H05

ASSESSORS:

Name	1.	Sifiso Nyambi
	2.	Maponya Khupo

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-001	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF MOISTURE CONTENT OF COAL IN THE ANALYSIS COAL SAMPLE		

Instructions and Information

1.	The competency-based assessment instrument was developed in line with the requirements to be declared competent in **the Determination of Inherent Moisture** **Desired Outcome:** At the end of this module the employee should be able to perform Determination of inherent Moisture in line with the test procedure in a manner which will give accurate, repeatable and traceable results. The employee should have adequate knowledge of the method and its limitations
2.	You are allowed to use the following documentation during the assessment: • Any manual(s) associated with Inherent Moisture determination(s) in use. • Any Corporate Standard, Corporate Guideline or Corporate Manual. • Any other relevant **printed** reference material you may feel will assist you.
3.	You are going to demonstrate the determination of the inherent moisture During the assessment, the assessor may ask you additional relevant questions, the answer to which he/she will then immediately note down on the assessment sheet. The assessment consists of **26** points of which **16** are critical (*). To be declared competent, you must obtain 80% for the assessment. You are not allowed to get one of the critical questions wrong. The assessment shall then be stopped, and the result will be that you are not yet competent. Some of the questions may be structured to assess your reflective competence.
4.	You have the right to an interpreter and the right to have an observer present during the assessment.
5.	You are allowed to ask the assessor(s) to clarify any question during the assessment.

AzurESky	AZURESKY PTY LTD	Document # AZCA-001	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF MOISTURE CONTENT OF COAL IN THE ANALYSIS COAL SAMPLE		

Assessment Inherent Moisture

96%

Date: 15/08

Candidate name: Abenathi Staff No: _____

Candidate Signature: Same

No	Question:	Points	Score
1.	At what temperature is the test sample analysed? 105°C – 110°C ✓	* 1	1
2.	Who is responsible for plotting control charts? Operator/analyst ✓	* 1	1
3.	Who is responsible for investigating any deviation? Operator ✓	* 1	1
4.	What is meant by the term repeatability? Analysis done ✓ by the same operator, using same apparatus under same condition	2	2
5.	What does high inherent moisture content indicate? low rank Coal ✓	1	1
6.	Complete the sentence: Oxidized Coals become porous As Coals ✓ become porous, oxygen enters easily which ~~may~~ would lead to combustion, affect calorific value. basically it like sponge readily to absorb air	* 2	2
7.	Where must the crucibles be cooled and for how long? work bench top for 10 minute ✓ What about dessicator?	* 2	1
8.	What accompanies each batch of samples for inherent moisture? Control ✓	* 1	1
9.	How must the crucible lid be? the lid must be open, no crack ✓	1	1

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-001	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF MOISTURE CONTENT OF COAL IN THE ANALYSIS COAL SAMPLE		

10.	Name the one desiccant that can be used? Silica gel ✓	1	1
11.	The inherent moisture sample is incinerated for one hour. True or False? false ✓	* 1	1
12.	The inherent moisture crucible is weighed back while still hot. True or False? false ✓	* 1	1
13.	Inherent moisture crucibles are cooled with the lid open True or False? false ✓	* 1	1
14.	What is the function of silica gel in the dessicators? absorb moisture ✓	1	1
15.	How does one regenerate silica gel? by placing silica gel into the moisture oven over a period	1	1
16.	How many samples must be used in the repeat analyses? samples that need to be repeated in duplicate	1	
17.	Within which limits must mass be controlled? 1g ± 0.0010g 0.95 - 1.05 ✓	* 1	1
18.	What are the repeatability limits for inherent moisture? IM content ≤ 5%. RL of 0.10% ✓✓ IM content > 5%. RL of 0.15% ✓✓	* 4	4
19.	Inherent moisture can only be done when the balance has been verified True/ False ✓ True	* 1	1
20.	What is the relation between inherent moisture and volatile matter? Inherent moisture is needed for correction of volatile matter ✓	1	1
	Total:	26	25

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-001	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF MOISTURE CONTENT OF COAL IN THE ANALYSIS COAL SAMPLE		

Comments by Assessor(s): Got half the marks for a critical question, will need to repeat the assessment since this is a requirement. Otherwise, employee shows excellent understanding of this method.

Comments by Candidate:

Assessor Declaration:

I hereby state that the candidate has been assessed according to the requirements of this document and found to be *competent / not yet competent** **and** *may / may not* proceed* **to the Assessment Panel for authorization.**

Assessor Name Sifiso M **Signature:** _____ **Date:** 08/13/2025

Learner Signature: _____ Date: 13/08/25

Note: 1. This shall be done after theoretical test has been completed.
2. Trainee to be constantly supervised until Initial training is signed off by competent trainer.

Sample Used		Lims Number	Known Value	Duplicate Test Results	
Acceptance Tolerance: 0.15% between duplicate results for IM >5% and 0.1 for IM ≤5%					